



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
12-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

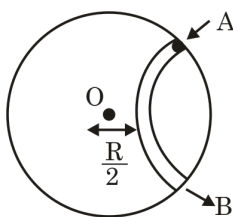
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Assume that there is a tunnel in the shape of a circular arc through the earth. Wall of the tunnel is smooth. A ball of mass 'm' is projected into the tunnel at A with speed 'v', the ball comes out of the tunnel at B and escapes out of the gravity of earth. Mass and radius of earth are M and R respectively and radius of circular shaped tunnel is also R.



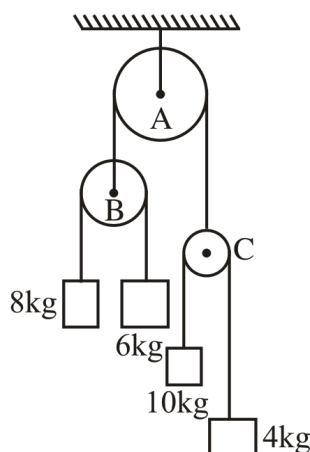
(A) The minimum possible value of 'v' is $v_0 = \sqrt{\frac{2GM}{R}}$.

(B) The minimum possible value of 'v' is $v_0 = \sqrt{\frac{GM}{2R}}$.

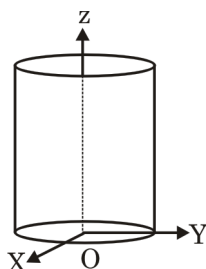
(C) If $v = v_0$, the normal force applied by the tunnel when it is closest to the centre of earth is $\frac{27GMm}{4R^2}$.

(D) If $v = v_0$, the normal force applied by the tunnel when it is closest to the centre of earth is $\frac{13GMm}{4R^2}$.

2. Consider the situation shown in the adjacent figure and select the correct statement(s) from the following. String and pulley both are ideal :-



- (A) Acceleration of pulley B is $\frac{g}{11}$.
- (B) Acceleration of pulley C is $\frac{g}{5}$.
- (C) Tension in the string passing over pulley A is 12.46 g.
- (D) Tension in the string passing over pulley A is 10 g.
3. The electric field in the space is given by $\vec{E} = E_0 (x\hat{i} + y\hat{j} + z\hat{k})$. Consider a right circular cylindrical surface whose radius is 'a' and height 'h'. Now choose the correct option(s) :-

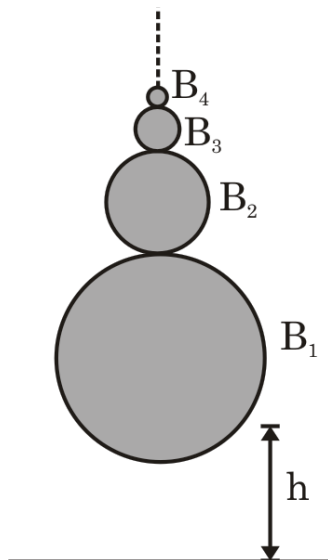


- (A) The electric flux through lower circular base is zero.
- (B) The electric flux through upper circular top is $\pi a^2 h E_0$.
- (C) The electric flux through lateral surface is $2\pi a^2 h E_0$
- (D) The total electric flux through cylindrical surface is $4\pi a^2 h E_0$.

Space for Rough Work

4. Consider n balls, $B_1, \dots, B_n$, having masses $m_1, m_2, \dots, m_n$ (with $m_1 \gg m_2 \gg \dots \gg m_n$), sitting in a vertical stack. The bottom of B_1 is a height h above the ground, and the bottom of B_n is a height $(h + \ell)$ above the ground. The balls are dropped from rest.

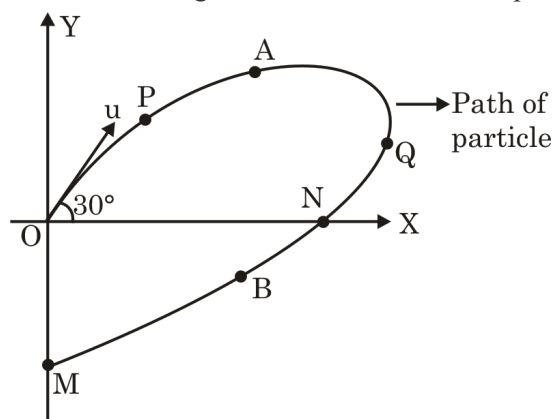
Work in the approximation where m_1 is much larger than m_2 , which is much larger than m_3 , etc, and assume that the balls bounce elastically. Also consider size of ball is much lesser than distance travelled by any of the balls. Then which of the following option(s) is/are correct ?



- (A) If $n = 2$, then the top ball bounces to a height $9h$ from ground.
(B) If $n = 4$, then the top ball bounces to a height $100h$ from ground.
(C) The maximum value of n for the top ball to reach atleast a height of $(1000)h$ from ground is 6.
(D) The value of n for the top ball to reach atleast a height of $(1000)h$ from ground is 4.

Space for Rough Work

5. A particle is projected from origin with speed $u = 50 \text{ m/s}$ at an angle 30° with X-axis. Due to wind particle acquires an extra acceleration $a = 6 \text{ m/s}^2$ in negative x-direction. Path of particle is shown in figure.



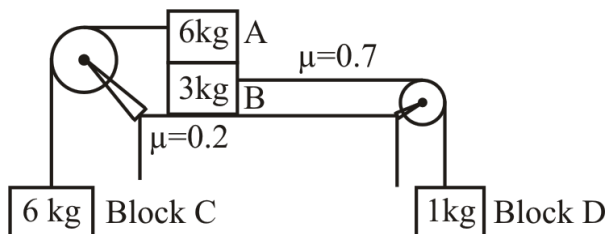
X-coordinates of point A and B are same.

Y-coordinates of P and Q are same.

Point N on the X-axis, point M on the Y-axis.

Time taken by particle from O to P, O to Q, O to A and O to B are t_1 , t_2 , t_3 and t_4 respectively (Assuming Y along vertical and there is no surface in the space) ($g = 10 \text{ m/s}^2$ and constant).

- (A) The value of $t_1 + t_2$ is 5 sec. (B) The value of $t_3 + t_4$ is 25 sec.
- (C) The value of ON is $125\sqrt{3}(5 - \sqrt{3})\text{m}$ (D) The length of OM is $\frac{625}{\sqrt{3}} \left(\frac{5}{\sqrt{3}} - 1 \right) \text{m}$.
6. An arrangement of masses and pulleys is shown in the figure. String connecting masses A and B with the pulleys are horizontal and all the pulleys and the string s are light. Friction coefficient between the surface and the block B is 0.2 and that between blocks A and B is 0.7. The system is released from rest. (use $g = 10 \text{ m/s}^2$)



- (A) The magnitude of acceleration of the system is 2 m/s^2 and there is no slipping between block A and block B.
- (B) The magnitude of friction between block A and block B is 42 N.
- (C) Acceleration of block C is 1 m/s^2 downwards.
- (D) Tension in the string connecting blocks B and D is 12 N.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

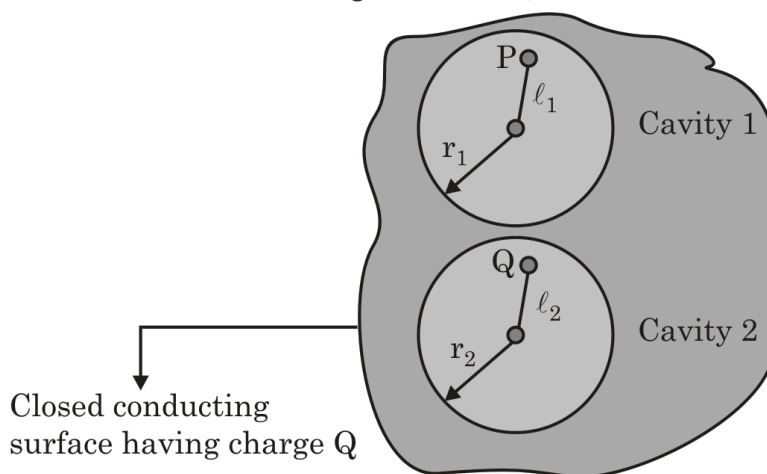
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

A point charge q_1 is placed inside the cavity 1 and another point charge q_2 is inside cavity 2. A point charge q is placed outside the conductor. For the given situation, answer the following questions :

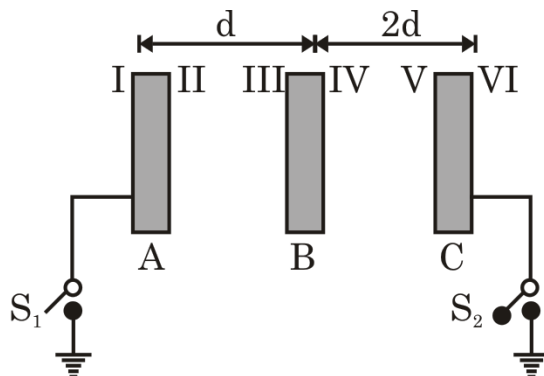


7. If q_1 is at centre of cavity 1, then E (electric field) at a point S at distance r from centre of cavity 1 ($r > r_1$) due to induced charge on the surface of cavity 1, is :-
- (A) $\frac{q_1}{4\pi\epsilon_0 r^2}$ away from the centre of cavity 1 (B) $\frac{q_1}{4\pi\epsilon_0 r_1^2}$ away from the centre of cavity 1
- (C) Zero (D) $\frac{q_1}{4\pi\epsilon_0 r^2}$ towards the centre of cavity 1
8. If the potential of the conductor is V_0 and charge q_2 is placed at centre of cavity 2, then potential at point Q , is :-
- (A) $\frac{q_2}{4\pi\epsilon_0 \ell_2} + V_0$ (B) $\frac{q_2}{4\pi\epsilon_0} \left(\frac{1}{\ell_2} + \frac{1}{r_2} \right) + V_0$
- (C) $\frac{q_2}{4\pi\epsilon_0} \left(\frac{1}{\ell_2} - \frac{1}{r_2} \right) + V_0$ (D) V_0

Space for Rough Work

Paragraph for Questions 9 and 10

Three large conducting plates are placed parallel to each other as shown in figure. The outer plates A and C are connected to ground with the help of switches as shown in figure. Initially, the plate B carries a charge $+Q$ while A and C are neutral and the switches are open. Based on above information answer the following questions.



9. When the switch S_1 is closed and S_2 is kept open, then :-
- (A) a charge of $+Q$ flows through S_1 from ground to plate A
- (B) a charge of $-\frac{Q}{2}$ flows through S_1 from ground to plate A.
- (C) the charge appearing on IV, V and VI surfaces are zero.
- (D) the charge appearing on surface II is zero.
10. When both the switches S_1 and S_2 are closed, then mark out the **INCORRECT** statement :-
- (A) Charge of $-\frac{2Q}{3}$ flows from ground to plate A.
- (B) Charge of $-\frac{Q}{2}$ flows from ground to plate A.
- (C) Charge on face V is $-\frac{Q}{3}$.
- (D) Charge on face III is $\frac{2Q}{3}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

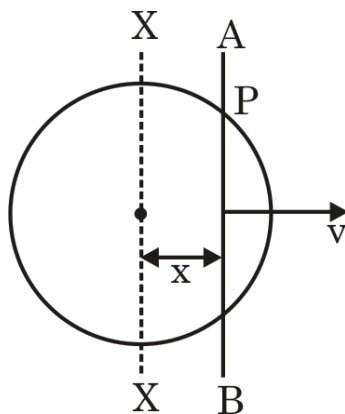
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A rod AB is moving on a fixed circle of radius R with constant velocity v as shown in the figure. Point P is the intersection of the rod and the circle.

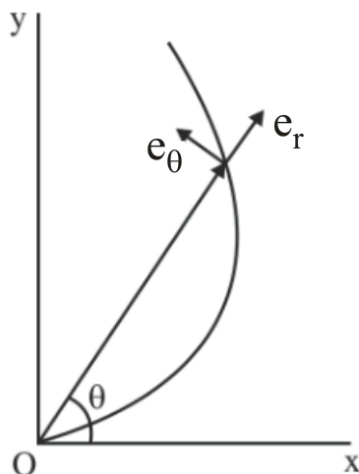


1. The velocity of point P at the instant when $x = \frac{3R}{5}$ is αv then value of α is
2. The angular velocity of point P at that instant is $\frac{5V}{\alpha R}$ then value of α is

Space for Rough Work

Paragraph for Questions 3 and 4

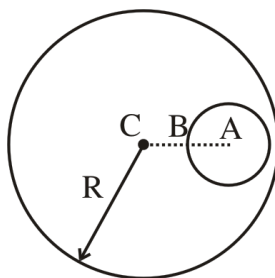
A particle moves along the path whose equation is $r = 2\theta$ m. The angle θ varies with time as $\theta = t^2$.



3. The velocity of particle when $\theta = 60^\circ$ is
4. The acceleration of particle when $\theta = 60^\circ$ is

Paragraph for Questions 5 and 6

A solid sphere of radius R has mass M and centre at C . A spherical cavity of radius $R/4$ and centre at A is cut as shown. Given $CB = BA = R/4$. Answer the following questions.



5. Gravitational potential at A is $\alpha \frac{GM}{R}$, the value of α is
6. Now a particle of mass $m (\ll M)$ is released from point A . The speed of particle when it strikes at B is $\alpha \sqrt{\frac{GM}{R}}$, the value of α is

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. A man throws a packet from the top of the tower directly aiming at his friend who is standing on the ground at a certain distance from the base of the tower and this distance is same as the height of the tower. If packet is thrown with a speed of 4 m/s and it hits the ground midway between the tower base & his friend, then height of the tower (in cm) is ($g = 10 \text{ m/s}^2$)
8. Three particle A, B and C are initially at points $\left(-\frac{3}{2}, 3, 0\right)$, $\left(\frac{3}{2}, 3, 0\right)$ and $\left(0, 3, \frac{3\sqrt{3}}{2}\right)$ respectively. They start their motion at $t = 0$ with same constant speed 2 m/s. Particle A always heads towards particle B, particle B heads towards particle C and particle C heads towards particle A. At what time will the particles meet each other.
9. A circle of radius 2m is marked on upper surface of a horizontal board initially at rest. A small toy car starts from rest along the circle with a tangential acceleration of 0.25 m/s^2 . At the same instant board accelerates with 2.5 m/s^2 in vertically upward direction. The coefficient of friction between board and small toy car is 0.1. Let s be the distance traveled (in meter) by the particle on the board without sliding. The value of s^2 is. Take $g = 10 \text{ m/s}^2$.
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Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

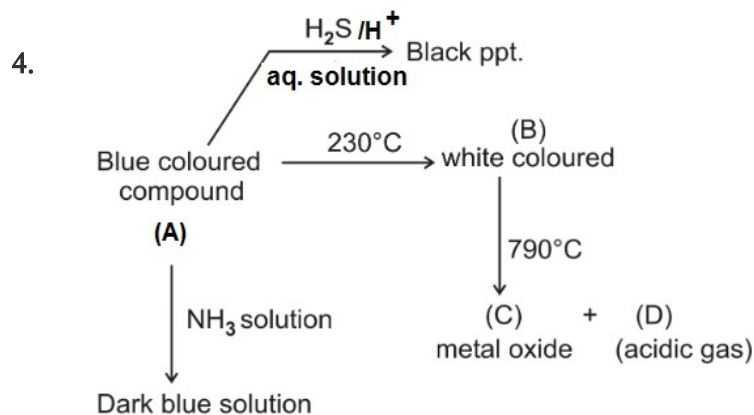
1. Which of the following statement/s correct about cement?
 - (A) Important ingredient present in port land cement are CaSiO_4 , Ca_3SiO_5 and $\text{Ca}_3\text{Al}_2\text{O}_6$
 - (B) The purpose of adding gypsum is only to slow down the process of setting of the cement so that it gets sufficient hardened
 - (C) For good quality cement, the ratio of SiO_2 to Al_2O_3 should be between 2.5 to 4
 - (D) When mixed with water the setting of cement takes place to give hard mass
2. In which of the following reaction/s H_2O_2 act as reducing agent?

(A) $\text{HOCl} + \text{H}_2\text{O}_2 \rightarrow$	(B) $\text{H}_2\text{O}_2 + \text{I}_2 \xrightarrow{\text{basic}}$
(C) $\text{Fe}^{3+} + \text{H}_2\text{O}_2 \xrightarrow{\text{basic}}$	(D) $\text{H}_2\text{O}_2 + \text{I}^- \xrightarrow{\text{acidic}}$

Space for Rough Work

3. Which of the following statement(s) is/are **CORRECT**?

- (A) All Cu(II) halides are known
(B) The standard reduction potential for Cu^{2+} to Cu has negative value
(C) Permanganate titrations in presence of HCl are unsatisfactory as HCl is oxidized to chlorine
(D) The E° value for $\text{Mn}^{3+}/\text{Mn}^{2+}$ is more positive than that of $\text{Cr}^{3+}/\text{Cr}^{2+}$ or $\text{Fe}^{3+}/\text{Fe}^{2+}$



Select correct statement(s) for the following?

- (A) (A) can form blue ppt. with NaOH
(B) In (A) 4 H_2O directly bonded with metal ion
(C) Black ppt insoluble in conc. HNO_3
(D) If dark blue colour is due to Switzer's salt then total three $2p\pi-3d\pi$ bonds present in one molecule of gas (D)

Space for Rough Work

5. Select the correct statement(s) :-
- (A) White ppt. of BaSO_3 and CaSO_3 is soluble in dil. HNO_3 , dil. HCl and CH_3COOH
 - (B) On standing, the precipitate BaSO_3 is slowly oxidised to sulphate which is insoluble in dilute mineral acids.
 - (C) Where excess of SO_2 gas is passed into the solution of BaSO_3 and CaSO_3 , then white turbidity disappears
 - (D) The hydrogen carbonates of alkali metals are soluble in water, but are less soluble than the corresponding normal carbonates
6. Which of the following is/are correct in the case of heavy water?
- (A) Heavy water is used as a moderator in nuclear reactor.
 - (B) Heavy water is more effective as solvent than ordinary water for ionic compound
 - (C) Heavy water has more boiling point than ordinary water
 - (D) Heavy water has lower boiling point than ordinary water

Space for Rough Work

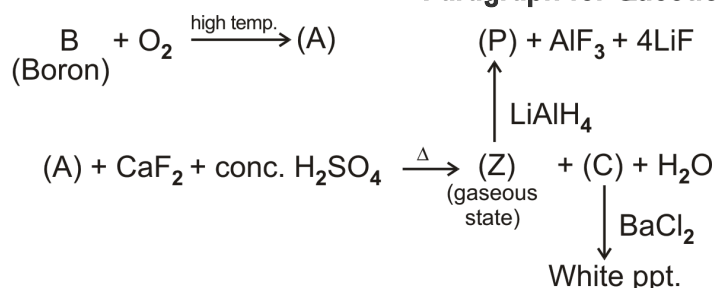
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Paragraph for Questions 7 and 8


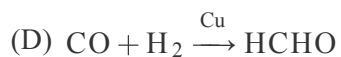
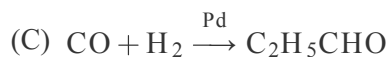
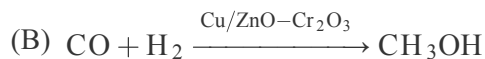
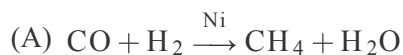
7. Which of the following more lewis acidic than Z molecule.
- (A) BI_3 (B) BCl_3 (C) BBr_3 (D) All of these
8. Which of the following give homolytic cleavage with compound (P).
- (A) CH_3NH_2 (B) CO (C) $(\text{CH}_3)_2\text{NH}$ (D) NH_3

Space for Rough Work

Paragraph for Questions 9 and 10

The mixture of CO and H₂ is called water gas mixture of CO and H₂ is used for synthesis of Methanol and a number of hydrocarbon, it is called Synthesis gas or Syn gas.

9. Which of the following reaction is incorrect?



10. The context with the industrial preparation of hydrogen from water gas (CO + H₂), which of the following is the correct statement?

(A) CO is removed by absorption in aqueous Cu₂Cl₂ solution

(B) H₂ is removed through occlusion with Pd

(C) CO is oxidized to CO₂ with steam in the presence of a catalyst followed by absorption of CO₂ in alkali

(D) CO and H₂ are fractionally separated using differences in their densities

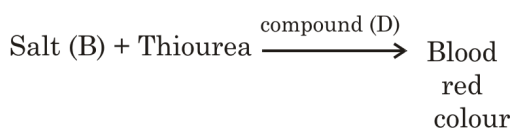
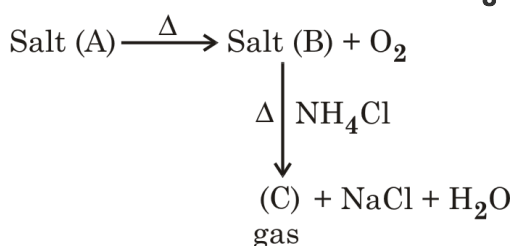
Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

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Paragraph for Questions 1 and 2


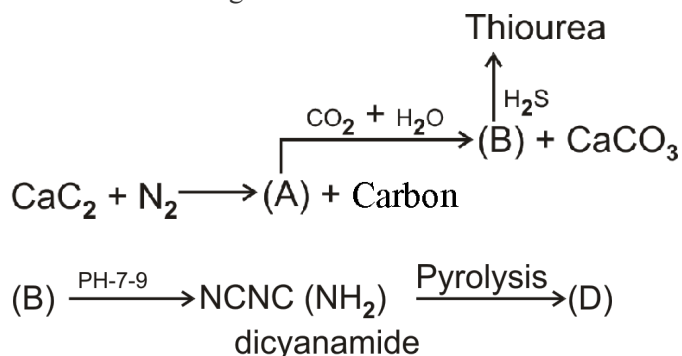
where C is diatomic inert gas which has highest concentration in atmosphere

- If x = Total number of element/s which react with gas (C) among following.
Li, Na, K, Rb, Cs, Be, Mg, Ca, Sr, Ba
 y = Total number of compound/s which give N_2 gas when react with salt (B) in acidic medium among following
 CH_3NH_2 , Urea, Thiourea, Sulphamic acid, Aniline (ice-cold condition)
Then calculate $(x - y)$
- Compound (D) + $\text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow \text{Compound (E)}$
(blue ppt.)
 $\text{H}_2\text{S} + \text{Compound (P)} + \text{compound (D)} \rightarrow \text{blue colour dye (Q)}$
if
 x = total number of ring/s present in blue colour compound (Q)
 y = total number of N atom present in compound (P)
then calculate $(x - y)$.

Space for Rough Work

Paragraph for Questions 3 and 4

Consider following reaction :-



Compound (D) + HCHO $\xrightarrow{\text{polymerisation}}$ Resin intermediate $\longrightarrow$ Melamine polymer

3. Total number of lone pair present in compound (D).
4. Total number of sp hybrid N atom present in compound (B).

Paragraph for Questions 5 and 6

Many d & f block ions are colored. The colour seems to depend on the many factors. Colours arises because light of a particular wavelength is absorbed in the visible region.

5. Total number of ions which are coloured in aq. solution.
La³⁺, Co²⁺, Fe³⁺, Ni²⁺, Lu³⁺, Gd³⁺, Ti⁴⁺, Zn²⁺
6. Total number of elements or ion/s which give H₂ with dil. HCl
Cu, Ti, Zn, Mn²⁺, Mn, Fe, Fe²⁺, Eu, Gd

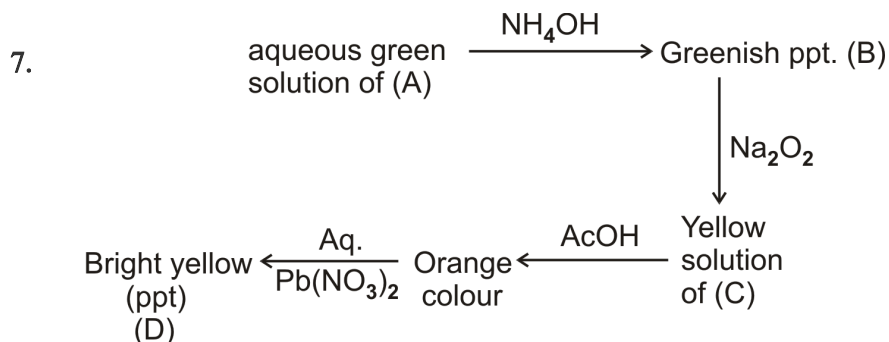
Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases



If x = oxidation state of central atom of (C) compound

y = number of d-orbitals used for hybridisation in compound (D)

then calculate value of $(x + y)$.

8. How many of the following reactions give yellow precipitate.
- $\text{NaBr} + \text{AgNO}_3 \rightarrow$
 - $\text{NaI} + \text{AgNO}_3 \rightarrow$
 - $\text{NaBr} + \text{Pb(NO}_3)_2 \rightarrow$
 - $\text{Na}_2\text{S} + \text{Cd(OAc)}_2 \rightarrow$
 - $\text{K}_2\text{CrO}_4 + \text{AgNO}_3 \rightarrow$
 - $\text{FeCl}_2 + \text{KCN (excess)} \rightarrow$
 - $\text{AgNO}_3 + \text{Na}_2\text{HPO}_4 \rightarrow$
 - $\text{FeCl}_3 + \text{CH}_3\text{COOH} \rightarrow$
9. When borax dissolves in water it form a complex compound, then what is the coordination number in that particular complex?

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If a pair of variable straight lines $x^2 + 4y^2 + \alpha xy = 0$ (where ' α ' is a real parameter) cuts the ellipse $x^2 + 4y^2 = 4$ at two points A and B , then the locus of the point intersection of tangents at A and B is...
 (A) $x - 2y = 0$ (B) $2x - y = 0$ (C) $x + 2y = 0$ (D) $2x + y = 0$
2. The series $\frac{8}{5} + \frac{16}{65} + \frac{24}{325} + \dots + \frac{8n}{4n^4 + 1}$
 (A) The sum to infinite number of terms of the series is 2
 (B) The sum to n terms of the series is less than 2, $\forall n \in \mathbb{N}$
 (C) The sum to n terms of the series cannot be an integer for any $n \in \mathbb{N}$
 (D) The sum to infinite number of terms of the series is $13/3$

Space for Rough Work

3. The locus of the image of the focus of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ with respect to any of the tangents to the ellipse is
- (A) $(x + 4)^2 + y^2 = 100$ (B) $(x + 2)^2 + y^2 = 50$
(C) $(x - 4)^2 + y^2 = 100$ (D) $(x - 2)^2 + y^2 = 50$
4. The locus of midpoint of the focal chord of a variable point moving on the parabola $y^2 = 4ax$ is a parabola whose
- (A) latus rectum is half the latus rectum of the original parabola
(B) vertex is $\left(\frac{a}{2}, 0\right)$
(C) directrix is y-axis
(D) focus has the co-ordinates $(a, 0)$
5. If $x + y = 2$ is a tangent to the ellipse whose foci are $(2, 3)$ & $(3, 5)$, then which of the following is/are correct ?
- (A) length of minor axis is 6 (B) length of minor axis is 3
(C) length of major axis is $\frac{\sqrt{41}}{2}$ (D) eccentricity of ellipse is $\frac{\sqrt{5}}{\sqrt{41}}$
6. If $\cos(\beta - \gamma) + \cos(\gamma - \alpha) + \cos(\alpha - \beta) = \frac{-3}{2}$, then $(\alpha, \beta, \gamma \in \mathbb{R})$
- (A) $\sum \cos \alpha = 0$ (B) $\sum \sin \alpha = 0$
(C) $\sin \alpha \sin \beta \sin \gamma = 0$ (D) $\sum (\cos \alpha + \sin \alpha) = 0$
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

If $a, b, c \in \mathbb{N}$ and the equations $x^2 - 5x + a = 0$, $x^2 - 6x + b = 0$, $x^2 - 8x + c = 0$ have exactly one root common and the common root is a natural number

7. Which of the following holds?

- (A) $3a = b + 2c$ (B) $3b = c + 2a$ (C) $3c = a + 2b$ (D) $a = b + c$

8. The common root is

- (A) 1 or 2 only (B) 3 or 4 only (C) 2 or 4 only (D) 1 or 2 or 3 or 4

Paragraph for Questions 9 and 10

Let x_1, x_2, x_3, x_4 be non zero real numbers. Let A, B, C, D be distinct points on the parabola $y^2 = 4x$. Suppose that $F(x_0, y_0)$ is midpoint of the focal chord AB and lines BC and AE are parallel where $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$, $D(x_4, y_4)$ and $E(2, 0)$

9. The value of y_3 is -

- (A) $4y_0$ (B) y_0 (C) $\frac{y_0}{2}$ (D) $2y_0$

10. If $y_1 y_4 = 4$ then the tangent at D and normal at A to the parabola meet at a point whose abscissa is -

- (A) $\frac{1 - x_0}{2}$ (B) $1 + x_0 - x_2$ (C) $1 - x_0 + x_2$ (D) $\frac{1 + x_0}{2}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let S_1 is a hyperbola and S_2 be an ellipse whose centers are at origin. The transverse axis of S_1 and major axis of S_2 lie along the x-axis and y-axis respectively. Let H be the hyperbola $x^2 - y^2 = 7$. The straight line $4x - 3y = 7$ touches the curves H, S_1 and S_2 at P, Q and R respectively. Suppose $PQ = \frac{5}{2} = QR$.

1. If e_1 and e_2 are eccentricities of S_1 and S_2 respectively. Then the value of $15e_1^2 + 4e_2^2$ is
 2. Number of common tangents in S_1 and S_2 is
-

Space for Rough Work

Paragraph for Questions 3 and 4

Let ABCD is a unit square and $0 < \alpha < 1$. Each side of the square is divided in the ratio $\alpha : 1 - \alpha$. These points are connected to obtain another square. The sides of new square are divided in the ratio $\alpha : 1 - \alpha$ and points are joined to obtain another square. The process is continued indefinitely. Let a_n denote the length of side and A_n be the n th square. (Including given Square)

3. If $\alpha = 1/3$, then the least value of n for which $A_n < 1/10$ is
4. The value of α , for which side of n^{th} square equals the diagonal of $(n+1)^{\text{th}}$ square, is

Paragraph for Questions 5 and 6

a, b, c, d are in A.P. and $d = a^2 + b^2 + c^2$.

(Where a, b, c and d are integers)

5. The common difference is
6. The value of $a + 2b + 3c + 4d$ is

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. If $\alpha = \frac{2\pi}{7}$, then the value of $\tan \alpha \tan 2\alpha + \tan 2\alpha \tan 4\alpha + \tan 4\alpha \tan \alpha$ is k, where $|k| =$
8. The length of the focal chord of the parabola $x^2 = 4y$, which touches the hyperbola $x^2 - 4y^2 = 1$, is equal to
9. From a point $(2, \alpha)$ tangents are drawn to the same branch of hyperbola $\frac{x^2}{100} - \frac{y^2}{16} = 1$. If range of $\alpha \in (\ell_1, \ell_2)$, then $[\ell_2] =$ (where $[.]$ denotes greatest integer function)
-

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work